An improved random AKS-class primality proving algorithm

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Abstract

We present an improved AKS condition $\binom{e\cdot|S|+de-1}{de-1} \geq n^{\lceil\sqrt{de/3}\rceil}$ for the random AKS algorithm, where $|S|$ is the number of congruences to be tested, e the degree of the modulo polynomial $x^e - r$ and d the multiplicative order of n modulo e. It is based on Bernstein's result and better than his condition $\binom{e\cdot|S|+e-1}{e-1} > n^{\lceil\sqrt{d^2e/3}\rceil}$ when $d > 1$; this improved condition enables us to choose a smaller e: theoretically by a factor $> d$ ($d \in (\log n)^{O(1)}$) and numerically $\geq d^2$ and $< d^3$ for most practical cases; and thus improves time and space complexities.

I. Introduction

A significant theoretical breakthrough in primality proving is the celebrated AKS algorithm, introduced by Agrawal, Kayal and Saxena in 2002 [1]. Its time complexity is $\tilde{O}(\log^{7.5} n)$. This result was subsequently improved to $\tilde{O}(\log^{6} n)$ by Lenstra and Pomerance [2]. These two algorithms are deterministic.

In 2002, Berrizbeitia proposed a probabilistic variant of the AKS-class algorithm with time complexity $(\log n)^{4+O(1)}$ [3]. His algorithm requires that $2^i \mid n^2 - 1$ where $2^i \approx \log^2 n$, and it was generalized by Cheng [4], Bernstein [5], Avanzi and Mihăilescu [6] in 2003. Cheng's generalization requires that $n-1$ has a prime divisor $e \approx \log^2 n$. But, as noted in [7, pp. 214-215], the number $n-1$ does not have such a prime or composite divisor e for some primes n. In order to solve this problem, Cheng resorted to the elliptic curve primality proving algorithm. Ezome and Lercier also followed this approach in 2013 [8].

On the other hand, Bernstein and Avanzi and Mihăilescu looked for a suitable divisor $e \approx \log^2 n$ of $n^d - 1$ for some small integer d. Bernstein proved that every prime n has a suitable pair (d, e); thus, solved the problem for the $n^d - 1$ approach [5]. However, parameter d appears only on the right-hand side of the AKS condition $\binom{e\cdot|S|+e-1}{e-1} \geq n^{\lceil\sqrt{d^2e/3}\rceil}$ in Theorem 3.2 [5]. Thus, a larger value of $|S|$ or e is required as d increases, which in turn leads to higher time (and space if e increases) complexities.

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In this paper, we apply the "smaller powers n^j" idea of Theorem 8.1 to Theorem 3.2 of [5], and give an improved AKS condition $\binom{e\cdot|S|+de-1}{de-1} \geq n^{\lceil\sqrt{de/3}\rceil}$. Compared with Theorem 3.2, this enables us to choose a smaller e when $d > 1$: theoretically by a factor $> d$ ($d \in (\log n)^{O(1)}$) and numerically $\geq d^2$ and $< d^3$ for most practical cases; thus, improves time and space complexities.

II. Main Theorem

Let $\mathrm{ord}_m(a)$ and $\mathrm{ord}_{\mathcal{R}}(a)$ denote the multiplicative order of a in rings $\mathbb{Z}_m$ and $\mathcal{R}$ respectively.

Theorem 1. *Let n, d and e be positive integers with $n > 1$ and $n^d - 1 = e \cdot U$ with $\gcd(e, U) = 1$. Let f be a monic polynomial in $\mathbb{Z}_n[y]$ of degree d. Define R as the ring $\mathbb{Z}_n[y]/(f)$. Let r be an element of R. Let S be a subset of $\mathbb{Z}_n$. Assume that*

C_1. $d = \mathrm{ord}_e(n)$;

C_2. $r^{n^d-1} = 1$ in R, $r^{(n^d-1)/q} - 1$ is a unit in R for each prime q dividing e;

C_3. $\binom{e\cdot|S|+de-1}{de-1} \geq n^{\lceil\sqrt{de/3}\rceil}$;

C_4. $\forall s, s' \in S$ ($s \neq s'$), s, $s^e - r$ and $s^e - (s')^e$ are units in R;

C_5. $(x - s)^{n^j} = x^{n^j} - s$ in the ring $R[x]/(x^e - r)$ for all $s \in S$ and $1 \leq j \leq d$.

Then n is a prime.

In condition C_5, we use d *manual* tests to obtain $(x-s)^{n^j} = x^{n^j} - s$ in $R[x]/(x^e - r)$ for $1 \leq j \leq d$. We will prove that this equation holds for all $j > 0$. In contrast, Bernstein tested only $(x-1)^n = x^n - 1$ in $R[x]/(x^e - y)$ in Theorem 8.1, and then he used the theoretical derivation to prove that $(x-1)^{n^j} = x^{n^j} - 1$ in $R[x]/(x^e - y)$ for all $j > 0$. Compared with Theorem 3.2, where only $(x - s)^{n^d} = x^{n^d} - s$ is tested in $R[x]/(x^e - r)$, Our d *manual* tests do not increase the time complexity since this test procedure, i.e., exponentiation $(x - s)^{n^j}$ for $j = 1, 2, \cdots, d$, is also used in Theorem 3.2.

Lemma 2. *Recall the notation in Theorem 1. We have $\mathrm{ord}_{e^2}(n) = d \cdot e$.*

Proof. Let $m = \mathrm{ord}_{e^2}(n)$, so $e^2 \mid n^m - 1$ and thus $e \mid n^m - 1$. Since $d = \mathrm{ord}_e(n)$, we thus have $d \mid m$. Write $m = d \cdot k$ for some positive integer k.

By $n^d - 1 = e \cdot U$, we have $n^{dk} = (n^d)^k = (1 + eU)^k = \sum_{j=0}^{k} \binom{k}{j}(eU)^j$.

For $j \geq 2$, the term $\binom{k}{j}(eU)^j$ is divisible by e^2; thus, congruent to 0 modulo e^2 and

$$n^{dk} = (1 + eU)^k \equiv \binom{k}{0} + \binom{k}{1}(eU) = 1 + keU \pmod{e^2}.$$

Since $m = dk = \mathrm{ord}_{e^2}(n)$, we have $n^{dk} \equiv 1 \pmod{e^2}$, so $1 + keU \equiv n^{dk} \equiv 1 \pmod{e^2}$ and thus $keU \equiv 0 \pmod{e^2}$. Hence, $e \mid kU$. But $\gcd(U, e) = 1$, it follows that $e \mid k$. The smallest positive such k is $k = e$, and e indeed satisfies the order condition. So $\mathrm{ord}_{e^2}(n) = d \cdot k = d \cdot e$. ∎

The framework of the proof of Theorem 1 is similar to that of Theorem 3.2 in [5].

Proof. **Step 1: Move to a field.**

Let p be a prime factor of n. Find an irreducible polynomial g in $\mathbb{Z}_p[y]$ dividing the image of f. Then $k := \mathbb{Z}_p[y]/(g)$ is a finite field with $|k| = p^{\deg(g)}$ elements. Find an irreducible polynomial h in $k[x]$ dividing the image of $x^e - r$. Then $\mathbb{F} := k[x]/(h)$ is a finite field with $|\mathbb{F}| = p^{\deg(g)\cdot\deg(h)}$ elements.

Write $N = |R| = n^d$. Define ζ as the image of $r^{(N-1)/e}$ in k. Then $\mathrm{ord}_k(\zeta) = e$.

Since $x^N = x \cdot x^{N-1} = x(x^e)^{(N-1)/e} = \zeta x$ in $k[x]/(x^e - r)$ and $\zeta^N = \zeta\zeta^{N-1} = \zeta \cdot r^{(N-1)(N-1)/e} \overset{C_2}{=} \zeta$ by condition C_2, we have

$$x^{N^i} = \zeta^i x \tag{1}$$

in $k[x]/(x^e - r)$ for any integer $i \geq 0$.

By condition C_5 ($j = d$), we have $(x - s)^{n^d} = (x - s)^N \overset{C_5}{=} x^N - s \overset{(1)}{=} \zeta x - s$ in $k[x]/(x^e - r)$. Substitute $\zeta^i x$ for x, we have $(\zeta^i x - s)^N = \zeta^{i+1}x - s$ in $k[x]/((\zeta^i x)^e - r) = k[x]/(x^e - r)$ and

$$(\zeta^i x - s)^{N^u} = \zeta^{i+u}x - s \tag{2}$$

in $k[x]/(x^e - r)$ for any integer $u \geq 0$. Since $\mathrm{ord}_k(\zeta) = e$, equation (2) gives us $e - 2$ new equations ($i = 0,\ 1 < u < e$) about linear polynomials $(\zeta^u x - s)$'s from a single test in condition C_5 ($j = d$).

Now, for any positive integer $c = d \cdot u + j$, where $u \geq 0$ and $0 \leq j \leq d - 1$, we have $n^c = N^u n^j$ and

$$(\zeta^i x - s)^{n^c} = [(\zeta^i x - s)^{N^u}]^{n^j} \overset{(2)}{=} (\zeta^{i+u}x - s)^{n^j} \tag{3}$$

in $k[x]/(x^e - r)$.

Substitute $\zeta^{i+u}x$ for x in condition C_5 for $1 \leq j \leq d - 1$, we have

$$(\zeta^i x - s)^{n^c} \overset{(3)}{=} (\zeta^{i+u}x - s)^{n^j} \overset{C_5}{=} (\zeta^{i+u}x)^{n^j} - s \overset{(1)}{=} x^{N^{i+u}n^j} - s = (x^{N^i})^{n^c} - s \tag{4}$$

in $k[x]/((\zeta^{i+u}x)^e - r) = k[x]/(x^e - r)$.

We note that the purpose we test the first $d - 1$ equations manually in condition C_5 ($1 \leq j \leq d - 1$) is to obtain equation (4), and the last equation ($j = d$) in condition C_5 equation (2).

Field $\mathbb{F}$ is a finite field of character p. Thus, for any integer m, either positive or negative, we have

$$(\zeta^i x - s)^{n^c p^m} \overset{(1)}{=} [x^{N^i} - s]^{n^c p^m} \overset{(4)}{=} (x^{N^i \cdot n^c} - s)^{p^m} = [x^{n^c \cdot p^m}]^{N^i} - s \ \text{ in } \mathbb{F}. \tag{5}$$

We note that $s \in S \subset \mathbb{Z}_n$ and thus $s^{p^m} = s$. But $s \in S \subset R = \mathbb{Z}_n[y]/(f)$ in Theorem 3.2.

Step 2: Prove $|I| \geq de$.

Let $\delta = \mathrm{ord}_{\mathbb{F}}(x)$ be the order of the image of x in $\mathbb{F}^*$, and define I as the subgroup of $\mathbb{Z}_\delta^*$ generated by n and p; thus, $|I| = |\{x^{n^c p^m} \mid m \in \mathbb{Z},\ c \in \mathbb{Z} \text{ and } c \geq 0\}|$.

First, we prove that e^2 divides $\delta = \text{ord}_{\mathbb{F}}(x)$. Let $v = \text{ord}_{\mathbb{F}}(r)$ be the order of the image of r in $\mathbb{F}^*$. By condition C_2, i.e., $r^{n^d-1} = 1$ in R, we have $r^{n^d-1} = 1$ in k and thus in $\mathbb{F}$. Consequently, $v \mid (n^d - 1)$. Since a root of $x^e - r = 0$ is also a root of $(x^e)^v - 1 = r^v - 1 = 0$, the image of x in $\mathbb{F}^*$ satisfies $x^{e \cdot v} - 1 = 0$; thus, we have $\delta \mid (e \cdot v) \mid e \cdot (n^d - 1)$, or $\delta \mid (e^2 \cdot U)$ since $n^d - 1 = e \cdot U$.

However, for each prime q dividing e, $r^{(n^d-1)/q} - 1$ is a unit in R and thus in $\mathbb{F}$ by condition C_2; thus, $x^{e \cdot (n^d-1)/q} - 1 = (x^e)^{(n^d-1)/q} - 1 = r^{(n^d-1)/q} - 1$ is a unit in $\mathbb{F}$, and $x^{e \cdot (n^d-1)/q} \neq 1$ in $\mathbb{F}$. Consequently, $\delta \nmid e(n^d - 1)/q$, or $\delta \nmid (\frac{e^2}{q} \cdot U)$ for each prime q dividing e. We thus have $e^2 \mid \delta \mid (e^2 \cdot U)$.

Now let $\delta = e^2 u$ where $u \mid U$, then $\text{ord}_u(n) \mid d$ since $u \mid U \mid n^d - 1$. We have proved that $\text{ord}_{e^2}(n) = de$ in Lemma 2; thus, $\text{ord}_\delta(n) = \text{ord}_{e^2 \cdot u}(n) = \text{lcm}[\text{ord}_{e^2}(n), \text{ord}_u(n)] = \text{lcm}[de, \text{ord}_u(n)] = de$ and $|I| = \text{lcm}[\text{ord}_\delta(n), \text{ord}_\delta(p)] \geq \text{ord}_\delta(n) = de$.

Step 3: Combinatorially enumerate many group elements (an improved AKS lower bound).

In Theorem 3.2 (the case $c_- = 0$), the order e of ζ in $\mathbb{F}^*$ is used to prove that any two distinct polynomials of degree less than e, which are products of a subset of $e \cdot |S|$ linear polynomials $(\zeta^i x - s)$ $(0 \leq i < e, \ s \in S)$, map to two distinct elements in $\mathbb{F}$.

We now give a better AKS low bound using δ and I. We will prove that any two distinct polynomials of degree less than $|I|$, which are also constructed using the same set of $e \cdot |S|$ linear polynomials $(\zeta^i x - s)$ $(0 \leq i < e, \ s \in S)$ as those in Theorem 3.2, will map to two distinct elements in $\mathbb{F}$. Because $|I| \geq de > e$ when $d > 1$, we obtain more elements of $\mathbb{F}^*$, or a better AKS lower bound.

Consider functions $a : \{0, 1, \cdots, e-1\} \times S \to \mathbb{Z}$ such that $\sum_{i,s} a(i,s)[a(i,s) \geq 0] \leq |I| - 1$. There are $\binom{e \cdot |S| + |I| - 1}{|I| - 1}$ such functions. Note that $(\zeta^i x - s)$ is a unit in $\mathbb{F}^*$, and $\gcd((\zeta^{i_1} x - s_1), (\zeta^{i_2} x - s_2)) = 1$ in $k[x]$ unless $(i_1, s_1) = (i_2, s_2)$ by condition C_4 [5, p. 393]. We claim that the products $\prod_{i,s} (\zeta^i x - s)^{a(i,s)}$ are distinct in $\mathbb{F}^*$. Hence, there are $\binom{e \cdot |S| + |I| - 1}{|I| - 1}$ such elements in $\mathbb{F}^*$. Note that functions a in Theorem 3.2 are: $a : \{0, 1, \cdots, e-1\} \times S \to \mathbb{Z}$ such that $\sum_{i,s} a(i,s)[a(i,s) \geq 0] \leq e - 1$.

Indeed, assume that a, b are two such functions, and

$$A(x) := \prod_{i,s} (\zeta^i x - s)^{a(i,s)} \stackrel{(1)}{=} \prod_{i,s} [x^{N^i} - s]^{a(i,s)} \quad \text{and} \quad B(x) := \prod_{i,s} (\zeta^i x - s)^{b(i,s)} \tag{6}$$

are two elements in $\mathbb{F}^*$ such that $A = B$. So we have $A^{n^c p^m} = B^{n^c p^m}$ for any positive integer c and integer m. Since $A(x^{n^c p^m}) = A^{n^c p^m} = B^{n^c p^m} = B(x^{n^c p^m})$ by $[x^{N^i} - s]^{n^c p^m} = [x^{n^c \cdot p^m}]^{N^i} - s$ in (5), all $|I|$ distinct elements in $\{x^{n^c p^m} \mid m \in \mathbb{Z}, c \in \mathbb{Z} \text{ and } c \geq 0\} = \{x^i \mid i \in I\}$ are roots of polynomial $A(Y) - B(Y) \in \mathbb{F}[Y]$. But the degree of $A(Y) - B(Y)$ is $\leq |I| - 1$, a contradiction.

The reason for this improved AKS lower bound is that we use the "smaller powers n^j" in (4), we thus can use the *bigger* order $\delta = \text{ord}_{\mathbb{F}}(x)$: $e^2 \mid \delta$. While Theorem 3.2 uses the "bigger powers N^j", i.e., equation (2), and can only use the *smaller* order $e = \text{ord}_{\mathbb{F}}(\zeta)$. In fact, $\zeta = r^{(N-1)/e} = (x^e)^{(N-1)/e} =$

$x^{N-1} \in \langle x \rangle$ in $\mathbb{F}^*$, so $\langle \zeta \rangle$ is the unique subgroup of order e in $\langle x \rangle$, and ζ^i has less number of possibilities than that of x^i for integer $i > 0$.

Step 4: An improved AKS upper bound.

Recall that $\delta = \mathrm{ord}_{\mathbb{F}}(x)$ and I is the subgroup of $\mathbb{Z}_\delta^*$ generated by n and p.

The lattice $L := \{(\alpha, \beta) \in \mathbb{Z} \times \mathbb{Z} \mid (n/p)^\alpha p^\beta \equiv 1 \pmod{\delta}\}$ has determinant $|I|$.

Define C as the set of $(\alpha, \beta) \in \mathbb{R} \times \mathbb{R}$ such that

$$\max\{|\alpha| \lg(n/p), |\beta| \lg p, |\alpha \lg(n/p) + \beta \lg p|\} \leq \sqrt{|I|/3} \lg n.$$

If $n \neq p$ then C is a closed convex symmetric set of area $3(|I|/3)(\lg n)^2/[(\lg p)\lg(n/p)] \geq 4|I|$. By Minkowski's lattice point theorem [2], there is a nonzero point $(c, m) \in L \cap C$. Multiplying (c, m) by ± 1, we may assume that $c \geq 0$; thus, $(n/p)^c p^m \equiv 1 \pmod{\delta}$ and $x^{(n/p)^c p^m} = x$ in $\mathbb{F}$.

For $A = \prod_{i,s}(\zeta^i x - s)^{a(i,s)} \overset{(1)}{=} \prod_{i,s}[x^{N^i} - s]^{a(i,s)} \in \mathbb{F}$ constructed in (6), if $m \geq 0$ we have

$$A^{(n/p)^c p^m} = \prod_{i,s}[(\zeta^i x - s)^{n^c p^{m-c}}]^{a(i,s)} \overset{(5)}{=} \prod_{i,s}[(x^{n^c p^{m-c}})^{N^i} - s]^{a(i,s)} = \prod_{i,s}[x^{N^i} - s]^{a(i,s)} = A.$$

Thus, field element A is a root of the polynomial $Y^{(n/p)^c p^m} - 1$ in $\mathbb{F}[Y]$. The degree $(n/p)^c p^m$ of this polynomial is $\leq n^{\sqrt{|I|/3}}$; thus, the AKS upper bound is $n^{\sqrt{|I|/3}}$.

Similarly, if $m < 0$ then A is a root of $Y^{(n/p)^c} = Y^{p^{-m}}$ for $x^{(n/p)^c} = x^{p^{-m}}$ in $\mathbb{F}$. The degree of this polynomial is $\leq \max\{(n/p)^c, p^{-m}\} \leq n^{\sqrt{|I|/3}}$; thus, the AKS upper bound is also $n^{\sqrt{|I|/3}}$.

So we have a contradiction by condition C_3: $\binom{e\cdot|S|+|I|-1}{|I|-1} \geq n^{\lceil \sqrt{|I|/3} \rceil}$ when $|I| = de$. Thus, $n = p$.

We have proved that $|I| \geq de$ in **Step 1**. In fact, a larger value of $|I|$ satisfies the AKS condition $\binom{e\cdot|S|+|I|-1}{|I|-1} \geq n^{\lceil \sqrt{|I|/3} \rceil}$ more easily than a smaller one. The following result is used in [6] to estimate a combinatorial number: If $m > c > 10$ then $\binom{m+c\cdot m-1}{m} > (2.5(c+1))^m$. After ignoring constants 2.5 and 3, this AKS condition can be simplified as $(\frac{e\cdot|S|}{|I|} + 1)^{|I|} \geq n^{\sqrt{|I|}}$, or $|I| \cdot \log^2(\frac{e\cdot|S|}{|I|} + 1) \geq \log^2 n$. Thus, a larger value of $|I|$ satisfies the condition more easily than a smaller one.

Note that if $n = p^i$ $(i \geq 2)$ then L still has determinant $|I|$ and C is still a closed convex symmetric set of area $(\frac{i^2}{i-1}|I|) \geq 4|I|$. Hence, the above contradiction still exists. However, C is not closed if $n = p$. So it is not necessary to check that n is a power of a prime. $\square$

III. Comparisons and Conclusions

Multiplying in $R[x]/(x^e - r)$ takes time $\tilde{O}(de \log n)$ since its elements are bivariate polynomials of degree less than e in x and less than d in y [9, Corollary 8.28]. The exponentiation $(x - s)^n$ requires $\tilde{O}(\log n)$ multiplications in $R[x]/(x^e - r)$. Thus, the time complexity of our algorithm is $|S| \cdot d \cdot \tilde{O}(de \log^2 n)$, which is equal to that of Theorem 3.2 of [5]. Space complexity of these AKS-class algorithms are $\tilde{O}(de \log n)$ bits of memory.

In order to compare time and space complexities of Theorem 1 and Theorem 3.2, we define $s := |S|$, $B :=$ the degree e in Theorem 3.2 and $E :=$ the degree e in Theorem 1. Since time and space complexities of these two theorems are all proportional to $\tilde{O}(e)$, the ratio B/E means that time and space complexities of Theorem 3.2 are about B/E times those of Theorem 1 respectively.

Applying Stirling formula $i! \approx \sqrt{2\pi i}\left(\frac{i}{\mathbf{e}}\right)^i$, where $\mathbf{e} \approx 2.718$, to $\binom{E\cdot|S|+dE-1}{dE-1} \geq n^{\lceil\sqrt{dE/3}\rceil}$, we have the following expression which determines the expression of $\sqrt{E} = \sqrt{e}$ in Theorem 1:

$$\binom{(d+s)E-1}{dE-1} \approx \binom{(d+s)E}{dE} = \frac{[(d+s)E]!}{(dE)!(sE)!} \approx \frac{\left(\frac{(d+s)E}{\mathbf{e}}\right)^{(d+s)E}}{\left(\frac{dE}{\mathbf{e}}\right)^{dE}\cdot\left(\frac{sE}{\mathbf{e}}\right)^{sE}} = \left(\frac{(d+s)^{d+s}}{d^d s^s}\right)^E = n^{\sqrt{dE/3}}.$$

Take logarithm on both sides of the last "=", we have

$$\sqrt{E}\,[(d+s)\log(d+s) - d\log d - s\log s] = \frac{\sqrt{d}}{\sqrt{3}}\log n.$$

Similarly, the expression of $\sqrt{B} = \sqrt{e}$ in Theorem 3.2 can be derived from $\binom{B\cdot|S|+B-1}{B-1} \geq n^{\lceil\sqrt{d^2B/3}\rceil}$:

$$\sqrt{B}\,[(1+s)\log(1+s) - s\log s] = \frac{d}{\sqrt{3}}\log n.$$

We thus have

$$\frac{B}{E} = d\left(\frac{(d+s)\log(d+s) - d\log d - s\log s}{(1+s)\log(1+s) - s\log s}\right)^2. \tag{7}$$

Thus, Theorem 1 reduces the lower bound of e by a factor $> d$ ($\in (\log n)^{O(1)}$ [5]) when $d > 1$.

As noted in [4] and [6] etc., the major problem of AKS-class algorithms relies not in the time but in the $\tilde{O}(de\log n)$ bits of space requirements for practical purposes. On the other hand, time complexities of Theorem 1 and Theorem 3.2 are all $|S|\cdot d\cdot\tilde{O}(de\log^2 n)$. Thus, a smaller e is preferred, which in turn needs a larger $s = |S|$ in the AKS condition. Indeed, Bernstein had stated this viewpoint in [5]: "reducing e at the expense of $|S|$ often saves time even if it increases $e|S|$".

Avanzi and Mihăilescu showed that $d = 60$ suffices for testing all numbers with 41000 decimal digits [6]. Using the deterministic AKS algorithm, i.e., modulo polynomial $x^e - \mathbf{1}$, Brent tested the primality of $n = 10^{49} + 9$ with ($e = 491$, $s = 28801$) in 32 hours and $n = 10^{100} + 267$ with ($e = 3541$, $s = 58820$) in about 1 years (estimated) [10]. He also reported that "Bernstein had proved $2^{1024} + 643$ prime by his improved random AKS-class algorithm in 13 hours on an 800 Mhz PC", and noted that "Elliptic curve primality proving (ECPP) algorithm takes about 50 seconds (Morain, on a 450 Mhz PC). Thus, ECPP is about 1660 times faster than Bernstein for numbers of this size".

For maximal values of d and s of such sizes, we list values of B/E in (7) in the following table. Numerically speaking, Theorem 1 reduces the value of e by a factor $\geq d^2$ and $< d^3$ except pairs $(s,d) \in \{(1,\geq 2),(2,\geq 5),(4,\geq 20)\}$. In fact, our condition $\binom{e\cdot|S|+de-1}{de-1} \geq n^{\lceil\sqrt{de/3}\rceil}$ has been simplified as $de\log^2(\frac{|S|}{d}+1) > \log^2 n$ at the end of the proof. Similarly, the condition $\binom{e\cdot|S|+e-1}{e-1} \geq n^{\lceil\sqrt{d^2e/3}\rceil}$

in Theorem 3.2 can be simplified as $\frac{e}{d^2}\log^2(|S|+1) > \log^2 n$. This rough estimation also supports the above numerical result, i.e., the improvement factor between d^2 and d^3. Thus, we attain noticeable improvements in both time and space.

TABLE I

RATIO OF ESTIMATED TIME AND SPACE COMPLEXITIES OF THEOREM 3.2 AND THEOREM 1

$\frac{B}{E}$ d / s	2	3	4	5	6	7	8	9	10	12	20	30	60
1	3.8	7.9	13.0	19.0	25.7	33.1	41.0	49.5	58.4	77.6	168.2	304.6	812.9
2	4.2	9.3	16.0	24.0	33.3	43.6	54.9	67.1	80.2	108.5	246.4	460.5	1284.5
4	4.7	11.0	19.6	30.5	43.4	58.1	74.6	92.6	112.1	155.2	373.6	726.8	2145.8
8	5.1	12.6	23.7	38.1	55.6	76.3	99.8	126.2	155.1	220.6	569.4	1164.1	3692.9
16	5.5	14.2	27.7	45.9	68.9	96.7	129.1	166.0	207.5	303.3	845.7	1832.0	6346.1
64	6.0	16.9	34.7	60.3	94.2	137.0	189.0	250.5	321.7	494.0	1592.6	3894.2	16579.6
512	6.5	19.4	41.9	75.7	122.5	183.8	260.8	354.7	466.7	748.7	2772.7	7699.5	42215.5
1024	6.7	20.0	43.6	79.5	129.5	195.5	279.1	381.6	504.5	816.5	3109.8	8864.0	51308.2
8192	6.9	21.4	47.5	88.0	145.6	222.6	321.4	444.2	593.0	976.8	3929.8	11776.5	75540.0
32768	7.06	22.0	49.4	92.2	153.4	235.9	342.2	475.0	636.8	1056.7	4347.4	13291.1	88705.6
65536	7.11	22.3	50.2	93.9	156.7	241.5	351.0	488.1	655.4	1090.7	4526.8	13946.4	94486.9

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